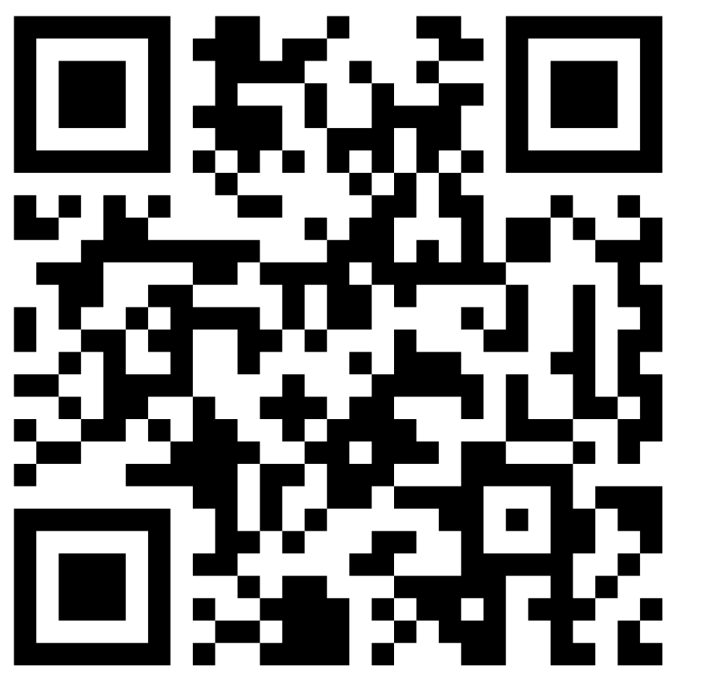


AdaBoosting Text Prompts for Vision-Language Models

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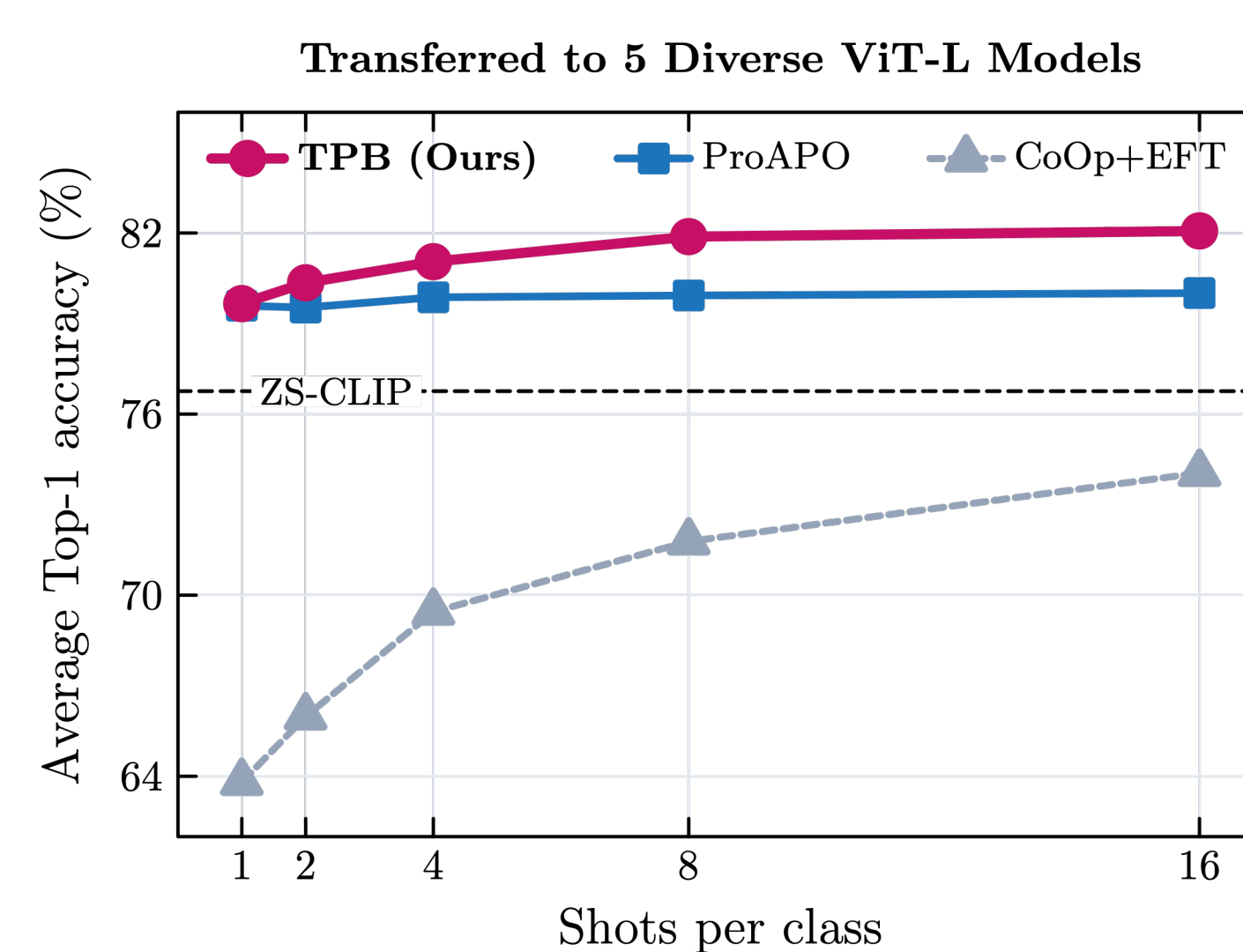
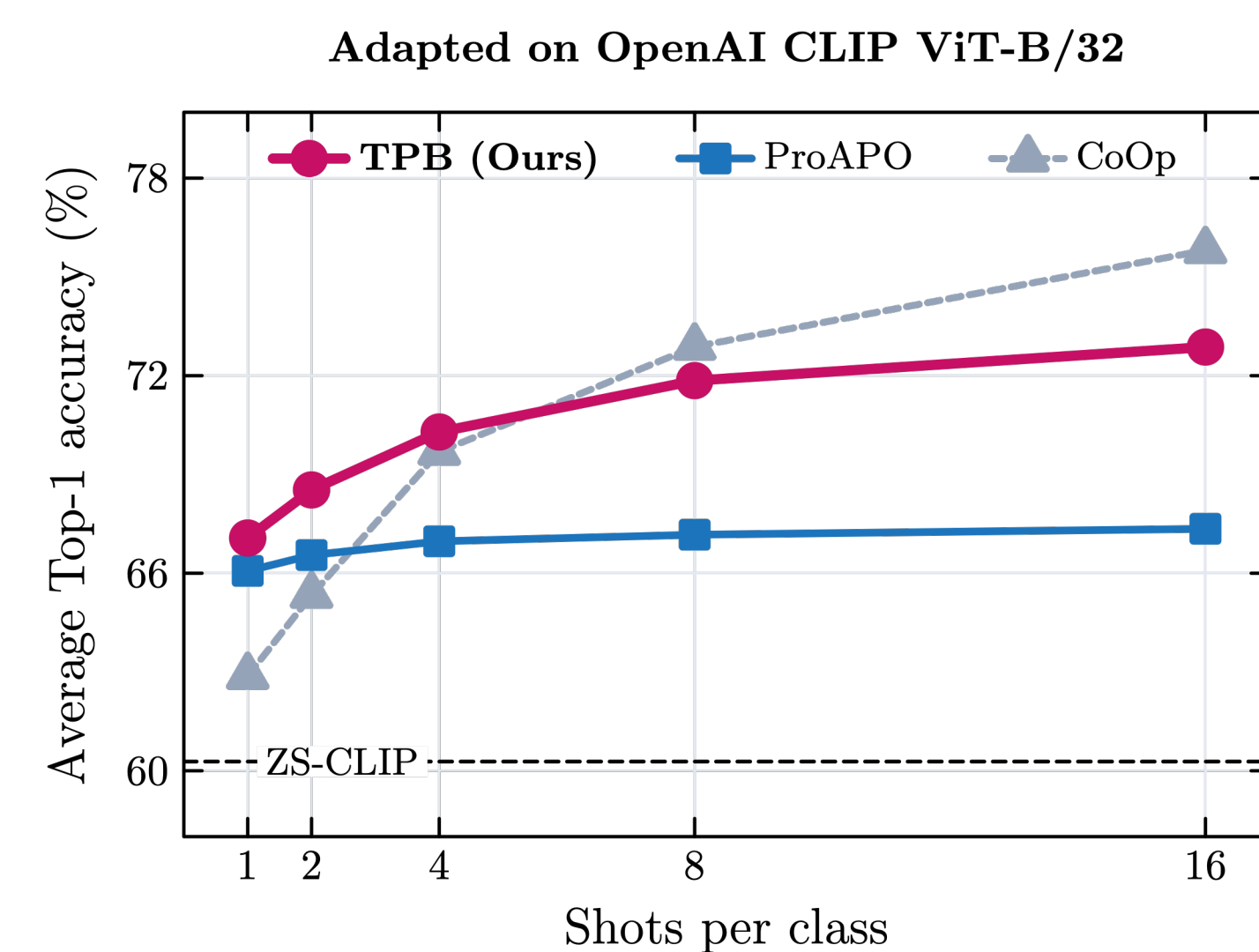


Overview

Prompt adaptation of VLMs faces a trade-off. **Soft prompts** improve with more shots but live in one model's embedding space; **hard prompts** transfer anywhere but stop improving.

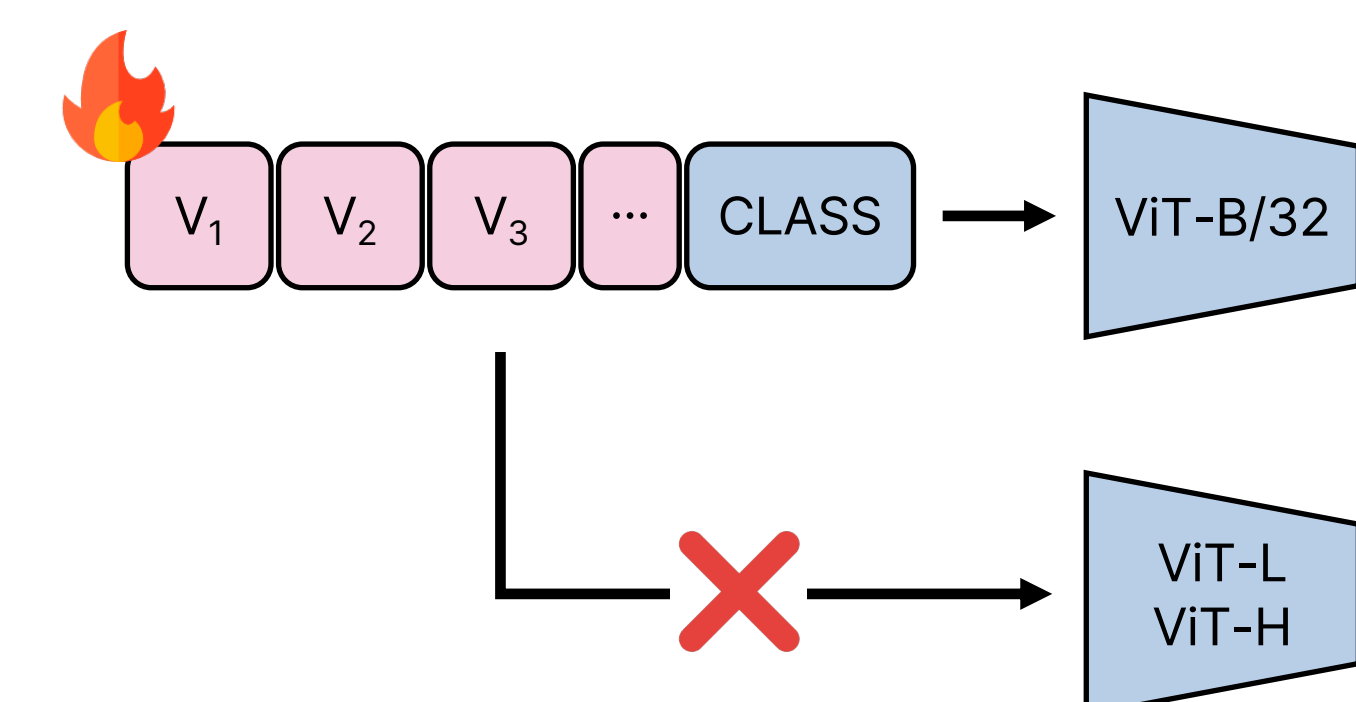
TPB (Text Prompt Boosting) breaks this trade-off:

- (i) **Shot scalability**: accuracy improves with more shots. **+5.80pp** from 1 to 16 shots on the source VLM.
- (ii) **Transfer robustness**: shot-driven gains preserve. **+2.41pp** on ViT-L and **+2.51pp** on ViT-H models.



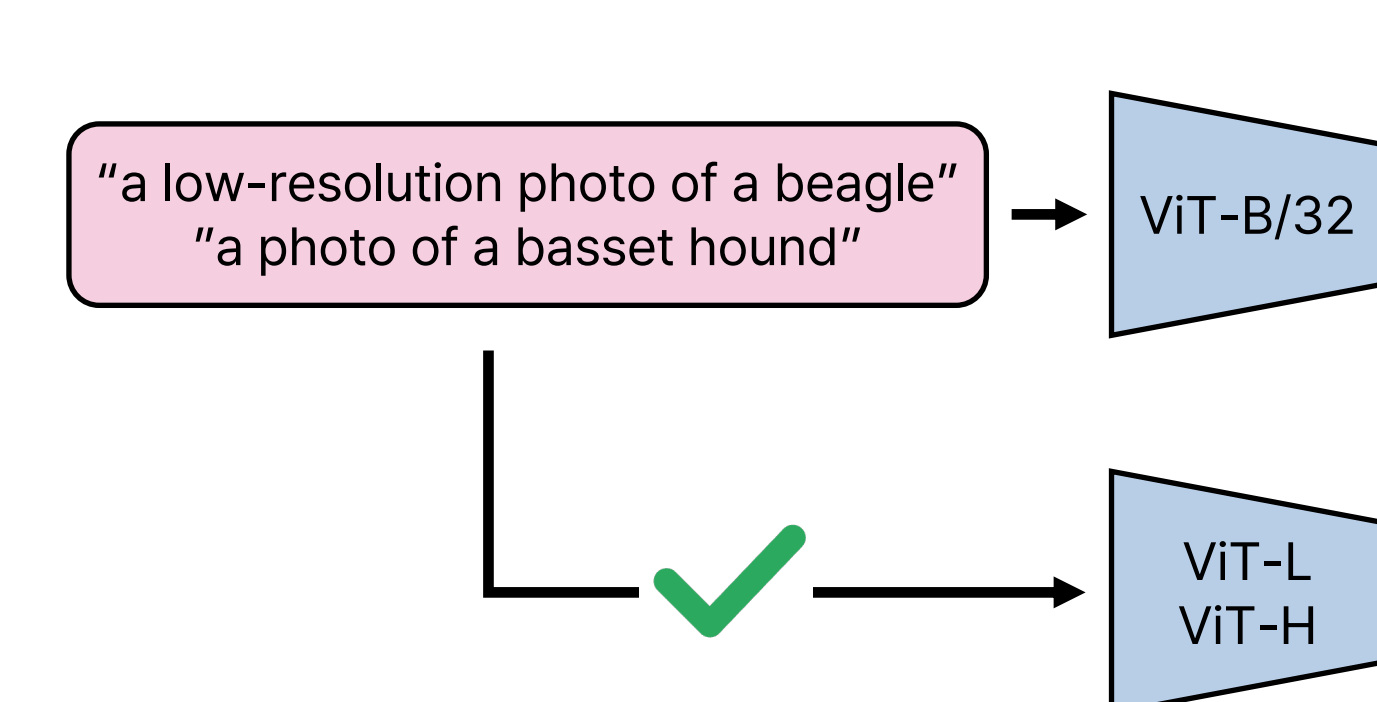
Why Boost Text Prompts?

1 Soft prompting



- ✓ Shot-scalable adaptation
- ✗ Direct cross-model transfer

2 Hard prompting



- ✗ Limited shot scalability
- ✓ Direct cross-model transfer

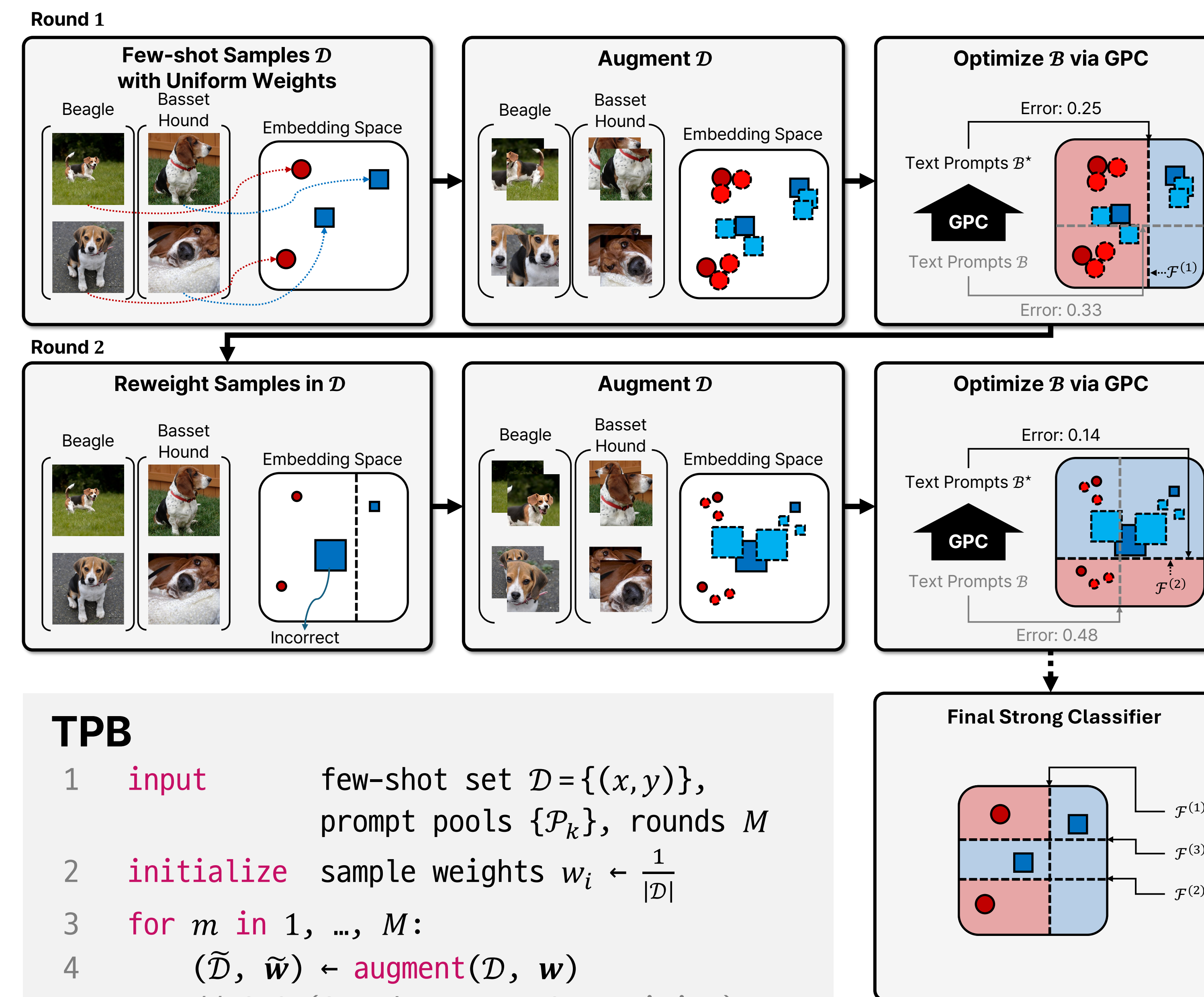
3 Text prompt boosting

Boosted text prompts scale with shots and transfer!

- **Compose**: greedily minimize weighted error.
- **Reweight**: focus subsequent learners on hard samples.
- **Ensemble**: combine weak classifiers into a strong classifier.

- ✓ Shot-scalable adaptation
- ✓ Direct cross-model transfer

Proposed Method



TPB

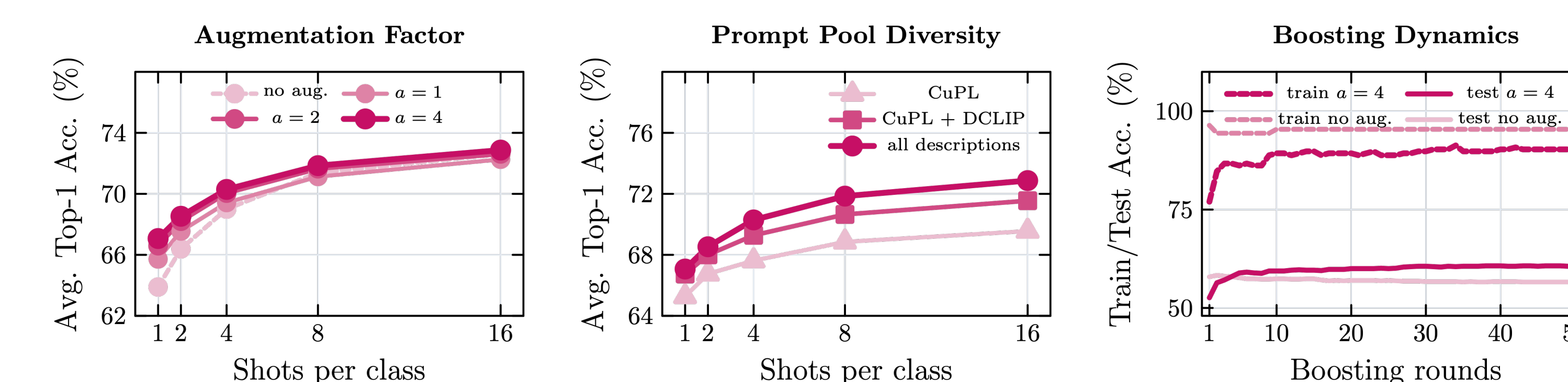
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1  input    few-shot set  $\mathcal{D}=\{(x,y)\}$ ,
           prompt pools  $\{\mathcal{P}_k\}$ , rounds  $M$ 
2  initialize sample weights  $w_i \leftarrow \frac{1}{|\mathcal{D}|}$ 
3  for  $m$  in  $1, \dots, M$ :
4       $(\tilde{\mathcal{D}}, \tilde{w}) \leftarrow \text{augment}(\mathcal{D}, w)$ 
           // GPC (Greedy Prompt Composition)
5       $\varphi^* \leftarrow$  best shared template on  $(\tilde{\mathcal{D}}, \tilde{w})$ 
6       $\mathcal{B}_k \leftarrow \{\varphi^*(k)\}$  for all class  $k$ 
7      repeat
8          greedily add class-wise prompts to reduce weighted error
9      until no improvement
10      $\mathcal{F}^{(m)} \leftarrow \text{classifier}(\{\mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_K\})$ 
           // Boosting
11      $w \leftarrow$  SAMME.R update( $\mathcal{F}^{(m)}, \mathcal{D}, w$ )
12  output    SAMME.R ensemble of  $\{\mathcal{F}^{(1)}, \dots, \mathcal{F}^{(M)}\}$ 
    
```

Analysis

Prompt diversity and augmentation improve adaptation, while boosting drives stronger cross-model transfer.

M : boosting round, a : augmentation factor



Fixed exposure budget $M \cdot a = 200$	Backbones		ZS-CLIP	M/a			
	Adapted on	OpenAI ViT-B/32		1/200	5/40	10/20	50/4
	Transferred to	ViT-L Models	60.10	74.42	72.94	73.32	73.36
		ViT-H Models	76.53	81.54	81.78	82.13	82.24
			78.60	83.12	83.72	83.98	84.49

Experimental Results

Shot scalability on the source model: RN50 & ViT-B/32

Top-1 Acc. (%), avg. of 11 datasets

Method	Adapted on OpenAI RN50					Adapted on OpenAI ViT-B/32				
	1	2	4	8	16	1	2	4	8	16
<i>Zero-shot methods</i>										
ZS-CLIP			55.98					60.28		
DCLIP			58.73					62.39		
CuPL			60.44					64.04		
<i>Few-shot methods</i>										
CoOp	59.34	61.73	66.51	69.93	72.98	62.88	65.34	69.69	72.88	75.81
PromptSRC	—	—	—	—	—	66.78	69.13	73.28	75.64	77.89
PEZ	42.36	48.96	53.87	58.37	61.09	46.26	53.12	57.37	61.34	64.20
ProAPO	62.65	62.66	63.44	63.93	64.39	66.07	66.55	66.97	67.17	67.35
TPB (Ours)	63.80	65.96	67.65	69.39	70.32	67.07	68.53	70.29	71.85	72.87

Cross-model transfer: ViT-B/32 → ViT-L & ViT-H models

Shot-driven gains preserve after transfer!

Top-1 Acc. (%), avg. of 11 datasets, ViT-L: OpenCLIP/SigLIP2/DFN/EVA-02/CLIPA-v2, ViT-H: OpenCLIP/DFN/CLIPA-v2/MetaCLIP2

Method	Transferred to ViT-L Models					Transferred to ViT-H Models				
	1	2	4	8	16	1	2	4	8	16
<i>Zero-shot methods</i>										
ZS-CLIP			76.76					78.75		
DCLIP			76.92					79.44		
CuPL			78.45					80.58		
<i>Few-shot methods</i>										
CoOp+EFT	63.81	65.98	69.45	71.78	74.05	62.69	64.63	68.17	70.40	72.73
PromptSRC+EFT	66.79	68.07	71.65	73.45	75.20	65.34	67.02	70.52	72.28	74.16
PEZ	49.05	53.21	55.56	57.84	58.93	46.97	51.50	54.59	56.47	57.60
ProAPO	79.59	79.53	79.87	79.93	80.01	82.07	81.90	82.26	82.25	82.39
TPB (Ours)	79.66	80.35	81.05	81.88	82.07	81.73	82.12	83.24	83.84	84.24

Gradient-Optimized "Text" ≠ Transferable

Gradient-optimized text prompts are still **"text"**, but they are **unreadable** and **fail to preserve** their gains under transfer.

Backbone	ZS-CLIP	PEZ	TPB (Ours)
RN50	55.98	61.09	70.32
ViT-L/14	71.07	53.02	78.46
Δ	+15.09	-8.07	+8.14

"hey darby pls violets kissestc
 accept liza any,, adorable
 gorgeous ♡♡ :-> life behaved pug"

Conclusion & Limitations

Conclusion: TPB builds and boosts an ensemble of text-prompt classifiers, yielding **shot-scalable gains** on the source model while **preserving these gains** across diverse VLMs.

Limitations: (i) Construction cost grows with dataset size. (ii) Reweighting may overemphasize noisy samples. (iii) Fixed prompt pool limit further adaption gains.